Open problems:
Unphysicality of Hilbert spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first. Items marked *research* are open research questions, not needed to check the brief.

From the brief

1. **Further problems.** *Research.* The list of ways in which QSTO:H fails the base conditions is open-ended: "If the reader is aware of other problems that should be added, do contact us." (Brief, Sec. 1, footnote 1.)

2. **A physically adequate model.** *Research.* The brief concludes that "the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements". A replacement for QSTO:H that satisfies [B]COBS, [B]WDQ and [B]STDIST is not given here; one candidate is the brief "Physical vector spaces for quantum mechanics". (Brief, Sec. 6.)

3. **Classical cutoffs.** For [B]COBS and [B]WDQ the brief says that in classical mechanics "one can, in principle, put a cutoff on all variables and work only with joint probability distributions in that compact region", which "bound all moments", while "Quantum mechanics does not allow that". Neither half is made precise. For the quantum half, one reading is that a nonzero wave function and its Fourier transform cannot both have compact support. (Brief, Sec. 4, proof of Theorem 5.)

The blocker of an earlier version of the brief, that the conditions and the inconsistency were informal, is resolved. The brief gives [B]COBS and [B]WDQ a minimum form (Sec. 3). Its current version states Theorem 5 as "Over QSTO:H, conditions [B]COBS, [B]WDQ and [B]STDIST are not guaranteed", which is shown by a condition failing for one particular system (Sec. 4). The current version also drops the minimum form of [B]STDIST, but the proof still fixes its meaning (see "Precise readings"). An earlier open problem, on the expectations of a general unbounded observable, is no longer raised: the [B]COBS part now rests on the number operator alone (Proposition 2). The formalization gap "Theorem 5 itself" is closed: Theorem 5 is stated and proved in Lean as `theorem5`, in the readings of the next section. The formalization gap "the Schrödinger equation with a potential" is closed too. The remark of Sec. 5, that the Schrödinger equation holds only on $\mathcal{D}(H)$, is checked for every self-adjoint realization H of $P^2/2m + V$, for every real V that is square integrable near each point of an open set with null complement (`realPotential_schrodinger_only_on_domain`), such as every continuous V or one continuous off 0; such an H exists by von Neumann's theorem. For bounded V and $V \in L^2 + L^\infty$, $H_0 + V$ is itself self-adjoint on $\mathcal{D}(H_0)$ (Kato–Rellich,

`memLp_schrodinger_only_on_domain`), and so is the only such H. These results are in modules of their own, outside the comparator lane, because TauCeti's theorems are reached through a private import: TauCeti and Physlib both declare `LinearPMap.adjoint_smul`. The former item "Schwartz spaces", on the closing remark of the proof of Theorem 5 that "Schwartz spaces present the same problem", is closed in the reading of the next section: for all $n, m \geq 1$ a unitary $L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^m)$ carries the QSTO:H model and restricts to an isomorphism of topological vector spaces $\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$ (`schwartz_stdist`, in a module of its own as well). No open problem now blocks verification.

Precise readings

The brief confirms the readings proposed for its earlier versions, with the changes below. They are the readings in which Theorem 5 is stated in Lean (`theorem5`).

Not guaranteed. "Over QSTO:H, condition C is not guaranteed" means: there is a quantum system to which QSTO:H applies whose QSTO:H model fails condition C. The current brief uses these words in place of "QSTO:H is inconsistent with C", which was read in the same way. The systems are the one-dimensional harmonic oscillator for [B]COBS, a system with one spatial degree of freedom for [B]WDQ, and two systems with different numbers of continuous degrees of freedom for [B]STDIST. Under this reading finite-dimensional Hilbert spaces do not need to be excluded.

[B]COBS.

- *States* are the nonzero vectors of $\mathcal{H} = \ell^2(\mathbb{N}_0)$, with the norm topology; this is equivalent to the quotient topology on $P(\mathcal{H})$, since everything below is invariant under $\psi \mapsto c\psi$.
- The observable is the number operator N with domain $\mathcal{D}(N)$ of Proposition 2.
- The *expectation* of N is $E_N(x) = \langle x|Nx\rangle / \|x\|^2$, defined for $x \in \mathcal{D}(N) \setminus \{0\}$.

B COBS, minimum form: E_N is continuous on $\mathcal{D}(N) \setminus \{0\}$ with the subspace topology.

- Claim to be proved: E_N is not continuous at e_0 within $\mathcal{D}(N) \setminus \{0\}$ (Proposition 2; `expN_not_continuousWithinAt` and `not_continuousOn_expN`, checked). That N is self-adjoint on $\mathcal{D}(N)$ and that $E_N(x) = \langle x|Nx\rangle / \|x\|^2$ there are checked too (`isSelfAdjoint_numberOp`, `coe_domain_numberOp`, `expN_eq_inner`).
- Not claimed by the brief, but true: for every densely defined, symmetric, unbounded O, $\psi \mapsto \langle O\rangle_\psi$ is discontinuous at every point of $\mathcal{D}(O) \setminus \{0\}$.

[B]WDQ.

- A *change of frame* is a smooth increasing bijection $y = f(x)$ of $\mathbb{R}$. It acts on states by the unitary $U\colon L^2(\mathbb{R}, dx) \to L^2(\mathbb{R}, dy)$, $(U\psi)(y) = \psi(g(y))\sqrt{g'(y)}$ with $g = f^{-1}$ (Proposition 3).
- Each observer measures position with the multiplication operator by their own coordinate, on its maximal domain: X on $L^2(\mathbb{R}, dx)$ and Y on $L^2(\mathbb{R}, dy)$. The brief now says this, and notes that UXU^{-1} is the different operator of multiplication by $g(y)$, with domain $U\,\mathcal{D}(X)$. It is checked that X and Y are self-adjoint (`isSelfAdjoint_positionOp`), that

UXU^{-1} is multiplication by $g(y)$ (`conjLinearPMap_frameUnitary_positionOp`) and self-adjoint (`isSelfAdjoint_conjLinearPMap_frameUnitary_positionOp`), and that $UXU^{-1} \neq Y$ (`conjLinearPMap_frameUnitary_positionOp_ne`).

- A normalized state has *finite variance* for an observable O if it lies in $\mathcal{D}(O)$, that is, if $\|O\psi\|^2 < \infty$ (Sec. 4).

B WDQ, minimum form: for every change of frame and every state ψ, $\psi \in \mathcal{D}(X)$ implies $U\psi \in \mathcal{D}(Y)$.

- Claim to be proved: for $f(x) = \tan(\frac{\pi}{2}\operatorname{erf}(x))$ and $\psi = (e^{-x^2}/\sqrt{\pi})^{1/2}$, $\psi \in \mathcal{D}(X)$ but $U\psi = (1/(\pi(1+y^2)))^{1/2} \notin \mathcal{D}(Y)$ (Proposition 3; `gaussState_mem_positionDomain`, `frameUnitary_gaussState`, `frameUnitary_gaussState_not_mem` and `not_forall_frameUnitary_mem_positionDomain`, checked).

[B]STDIST.

- The *systems* are those with $n \geq 1$ continuous degrees of freedom (Proposition 4 is stated for positive integers).
- The *mathematical representation* of a system is $\mathcal{H}_n = L^2(\mathbb{R}^n)$ with what QSTO:H builds from it: $P(\mathcal{H}_n)$, $D(\mathcal{H}_n)$, the self-adjoint operators, the Born rule and the von Neumann entropy. The brief now lists exactly these, so the positions $X_1, \ldots, X_n$ and momenta $P_1, \ldots, P_n$ are not part of the representation.
- Two systems have the *same* representation if a unitary $\mathcal{H}_n \to \mathcal{H}_m$ carries one onto the other.
- The current brief states [B]STDIST as "Systems with a different number of degrees of freedom are distinct", without saying in what sense. Its previous version had the minimum form "physical systems with a different number of degrees of freedom should not have the same mathematical representation". The proof of Theorem 5 still ends with the systems having "the same mathematical representation", so that minimum form is used.

B STDIST, minimum form: for $n \neq m$ there is no unitary $\mathcal{H}_n \to \mathcal{H}_m$.

- Claim to be proved: for all $n, m \geq 1$ there is one (Proposition 4; `L2_euclidean_iso`), and it carries the QSTO:H structure (`carriesQSTO`, `exists_carriesQSTO`). Both are checked.
- "Schwartz spaces present the same problem" is read as: the same holds for the rigged Hilbert spaces $\mathcal{S}(\mathbb{R}^n) \subseteq \mathcal{H}_n$. For all $n, m \geq 1$ there is a unitary $\mathcal{H}_n \to \mathcal{H}_m$ that carries the QSTO:H structure and maps $\mathcal{S}(\mathbb{R}^n)$ onto $\mathcal{S}(\mathbb{R}^m)$, as an isomorphism of topological vector spaces (`schwartz_stdist`, checked). The weaker reading, that $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^m)$ are isomorphic as topological vector spaces, follows (`nonempty_schwartzEquiv`).